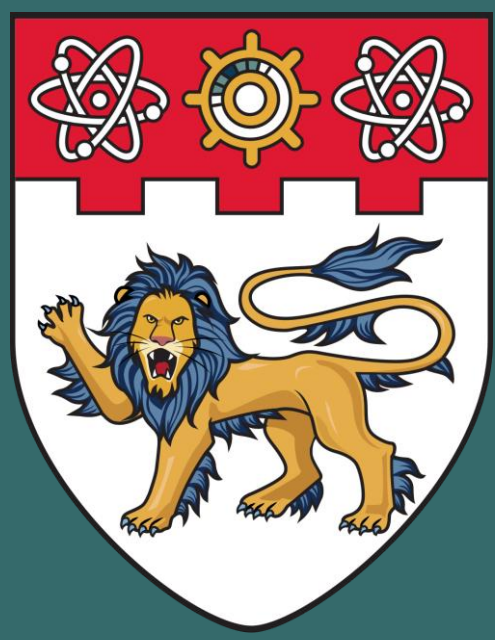




A Scalable HDX-MS Epitope-omics Workflow for Intact Dengue Virus at 90% Sequence Coverage for Structural Proteins

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INTRODUCTION

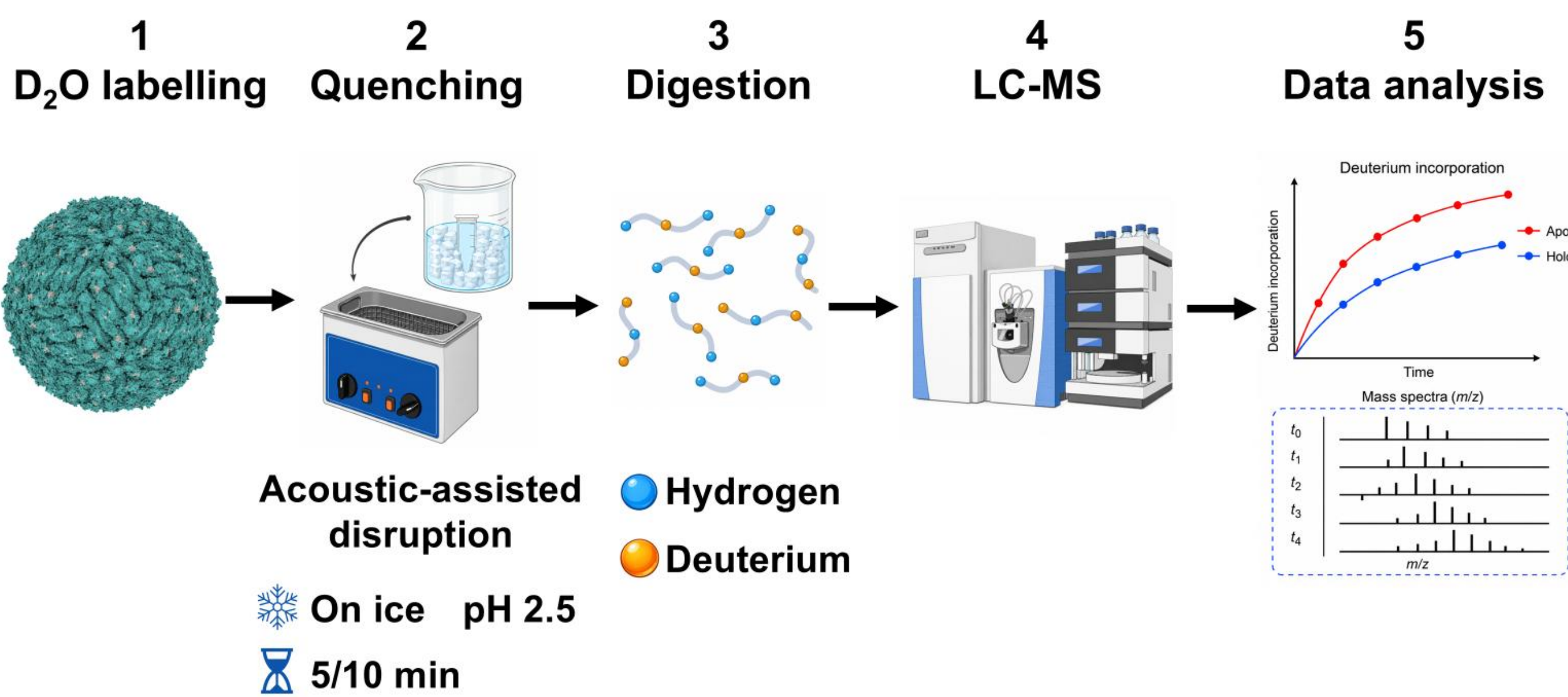
Epitope mapping is essential for understanding antibody–antigen interactions and supporting antibody development. Although X-ray crystallography, cryo-electron microscopy (cryo-EM) and nuclear magnetic resonance can provide high-resolution structural information, these methods can be expensive. Hydrogen-deuterium exchange mass spectrometry (HDX-MS) offers an alternative by measuring the rate of exchange protein backbone amide hydrogens with deuterium in the surrounding solvent, which can be used to probe protein dynamics, folding and solvent accessibility under near-native conditions. HDX-MS has also gained relevance in AI-assisted structural biology by providing experimental restraints for structural prediction¹. Many antibodies, including mAb 2D22, recognize quaternary epitopes, making the analysis of intact viral particles important for preserving native viral protein structure². Dengue virus serotype 2 (DENV2) presents a particularly challenging system, with 180 copies of the E and M proteins embedded within a lipid envelope and the C protein located beneath the surface layer³. Previous HDX-MS studies of intact DENV2 reported limited sequence coverage, particularly within membrane-associated regions⁴.

OBJECTIVES

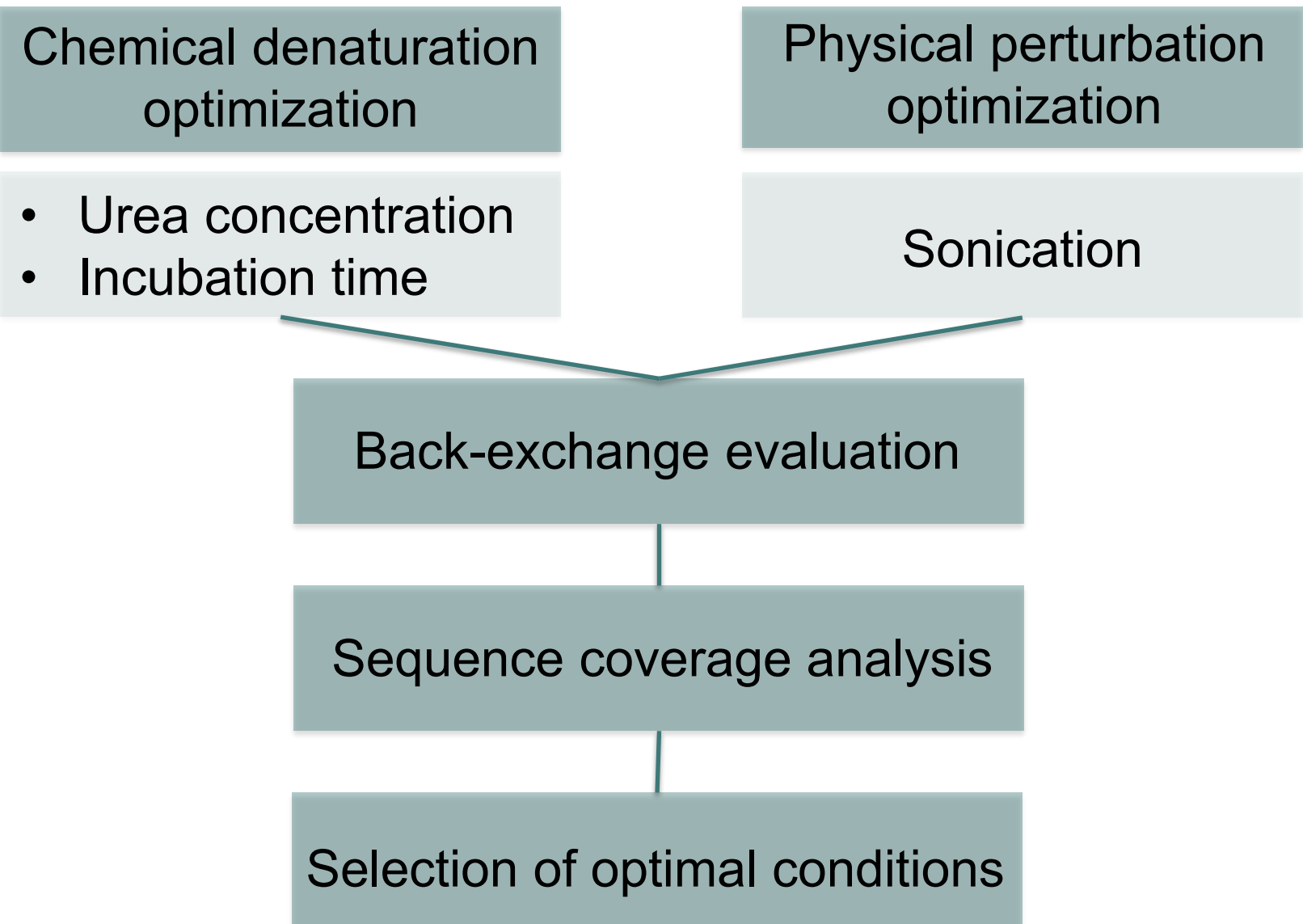
1. To optimize HDX-MS sequence coverage of DENV2 C, E and M proteins by evaluating chemical denaturation and physical perturbation during quenching.
2. To perform epitope mapping of mAb 2D22 on DENV2 D2Y98P E protein using HDX-MS.
3. To validate HDX-MS epitope mapping through comparison with cryo-EM structural data.

METHODS

Acoustic-assisted HDX-MS workflow



Acoustic-assisted HDX-MS method evaluation



CONCLUSION

1. Back-exchange was ~50% for Inc-5/Son-5 and ~55% for Inc-10/Son-10, within the guideline range of 10-55%⁵, with no apparent increase from sonication, indicating no disruption introduced by sonication.
2. Final quenching with 3 M urea with Son-10 increased sequence coverage to 96.0 ± 0.7%, 89.8 ± 0.8% and 87.1 ± 1.0% for the DENV2 D2Y98P C, E and M proteins respectively, achieving ~90% structural proteins coverage for reliable topology identification.
3. The resulting coverage enabled direct mapping of the 2D22 quaternary epitope on intact D2Y98P virions, with the regions identified at peptide-level resolution showing strong agreement with epitopes identified by cryo-EM.

ACKNOWLEDGEMENT

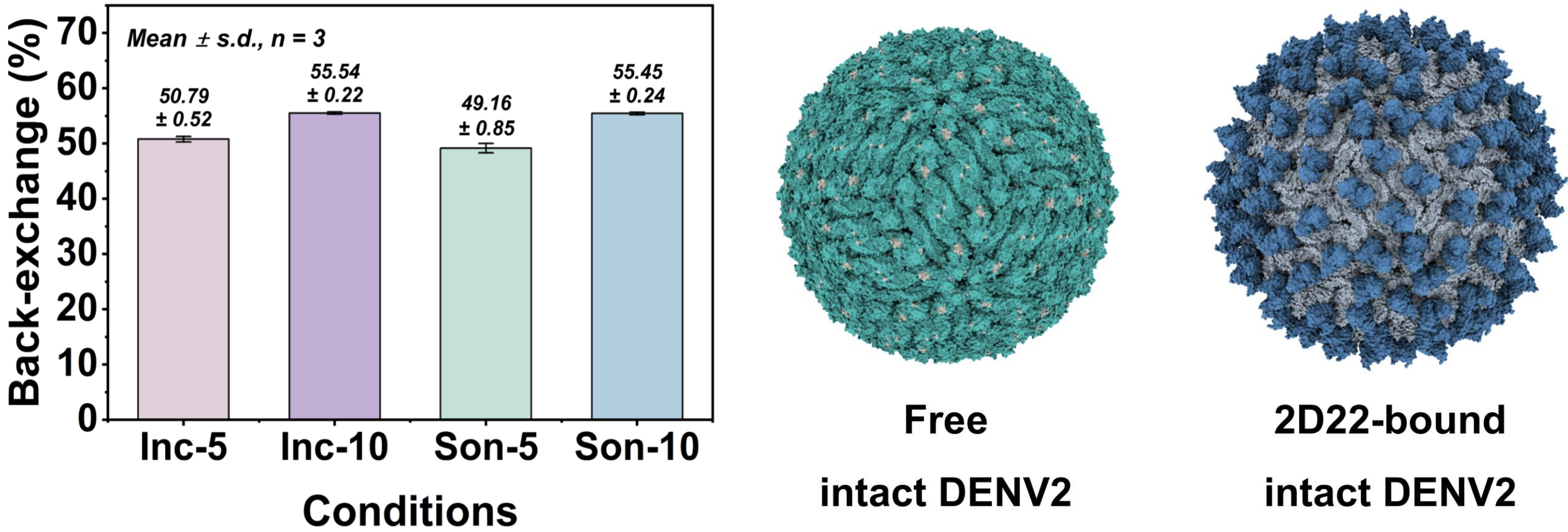
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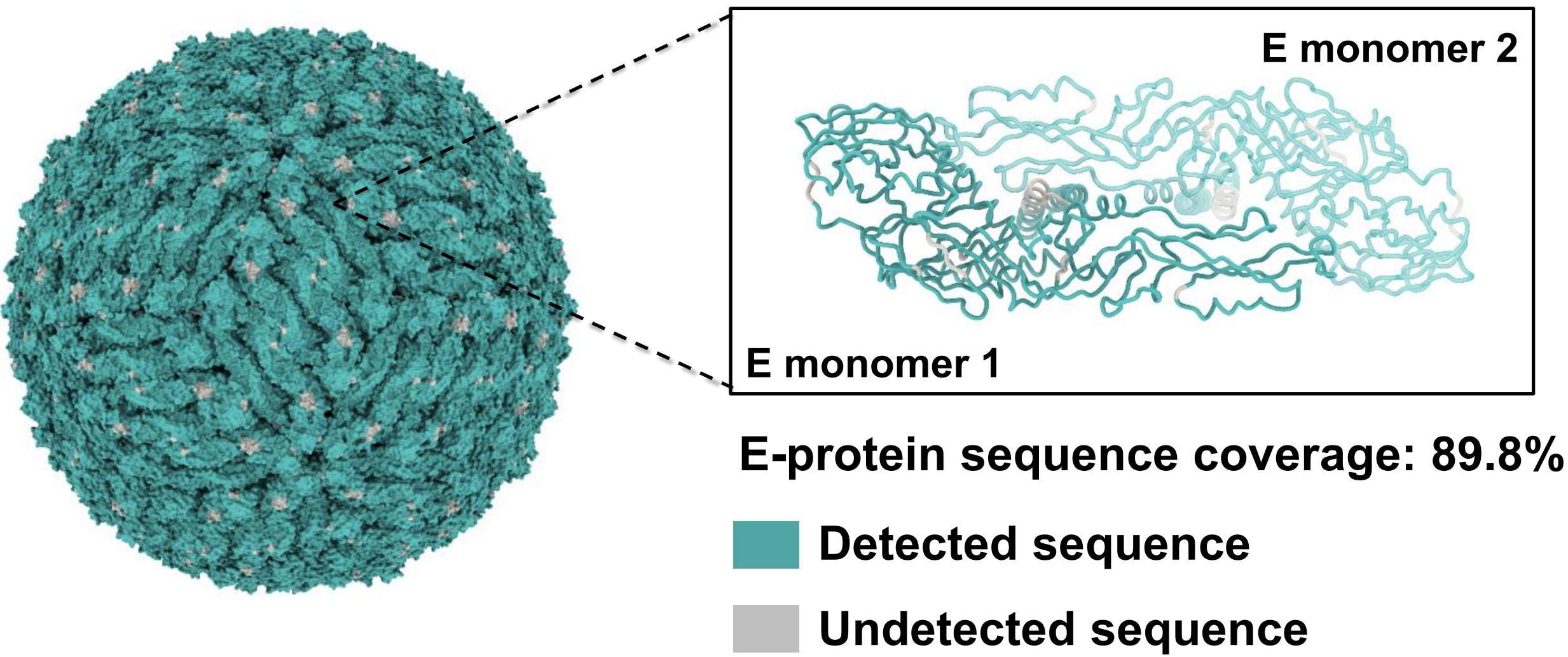
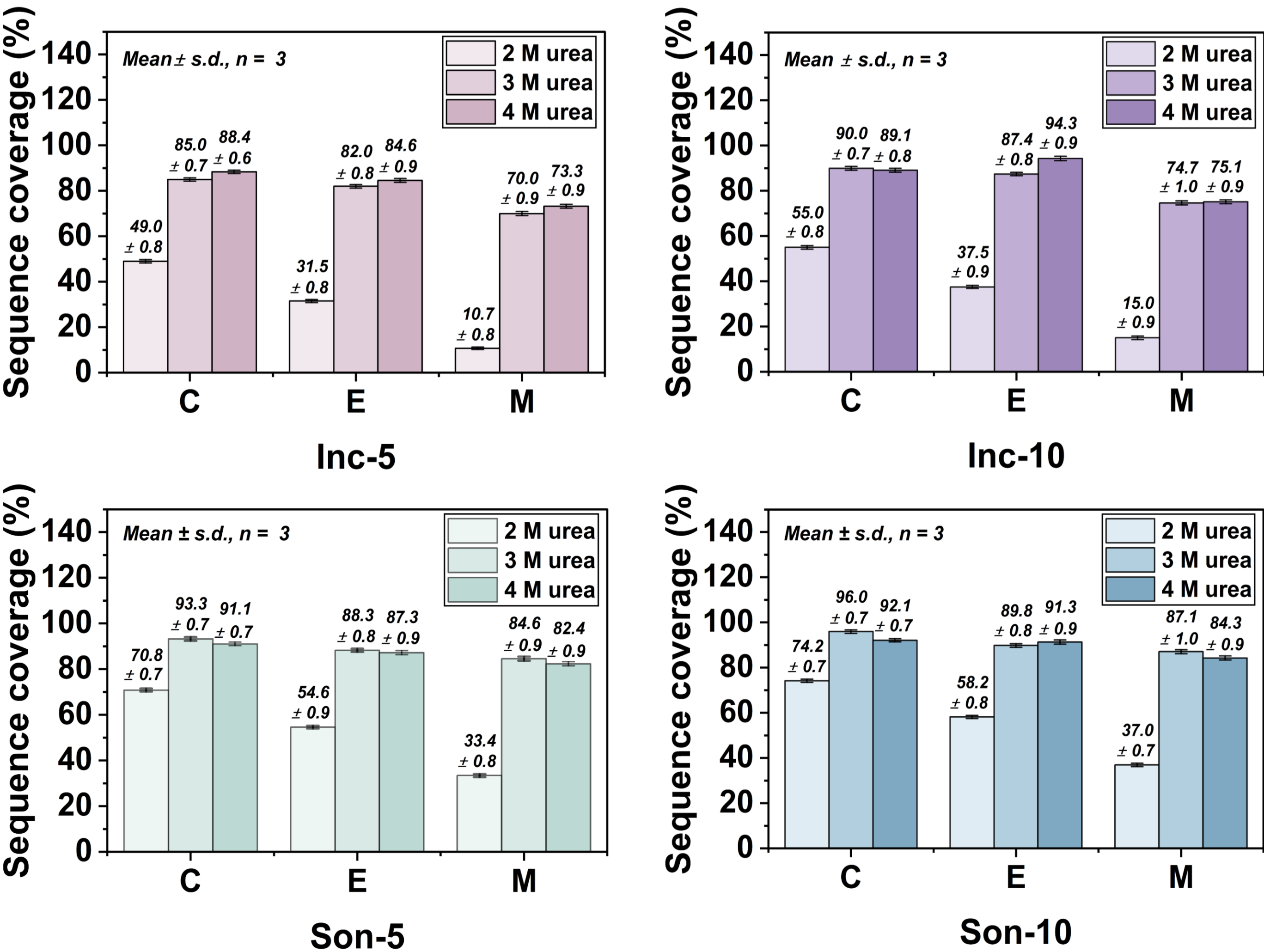
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RESULTS

Evaluation of back-exchange



Sequence coverage of intact viral proteins



HDX-MS epitope mapping for 2D22 on DENV2 D2Y98P and validation by cryo-EM

